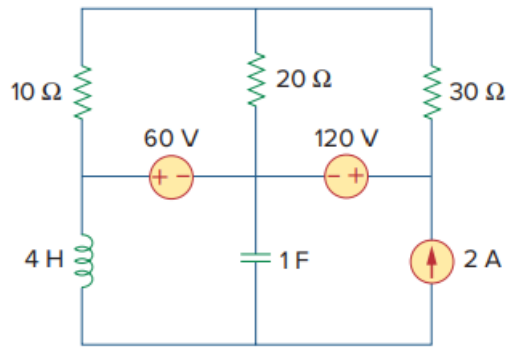


Problem

Find the dual of the circuit in Fig. 8.120.



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

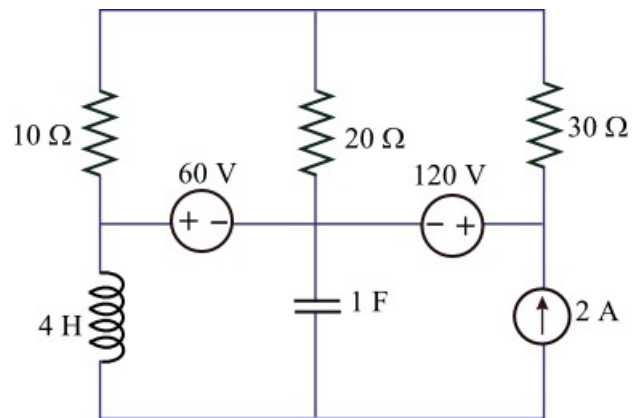


Figure 1

Step 2 of 5

Steps to follow to draw the dual of the circuit in Figure 1 are:

Step 1:

First locate nodes at the center of the each mesh and a reference node as; node 1, 2, 3, and 4 for four meshes and also the ground node 0 for the dual circuit as shown in Figure 2.

Redraw the circuit in Figure 1 by placing nodes:

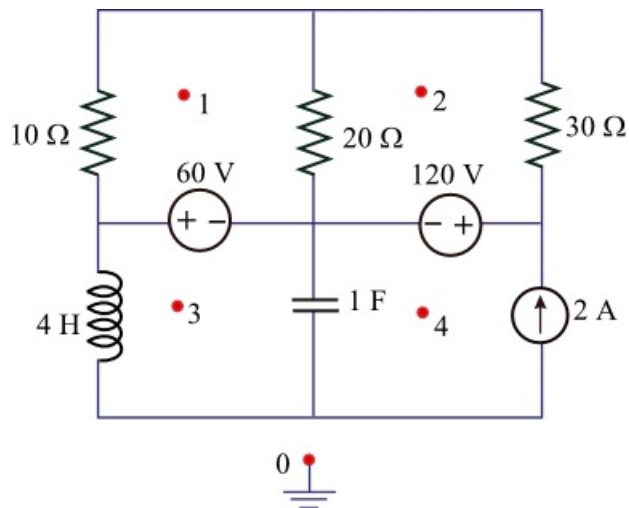


Figure 2

Step2:

Draw the lines between the nodes as shown in Figure 3 such that each line crosses an element. Replace the element by its dual;

- o Replace a line between the nodes 1 and 2 crossing the $20\ \Omega$ resistor by its dual $0.05\ \text{S}$ conductor.
- o Replace a line between the nodes 1 and 0 crossing a $10\ \Omega$ resistor by its dual $0.1\ \text{S}$ conductor.
- o Replace a line between the nodes 1 and 3 crossing the $60\ \text{V}$ voltage source by its dual $60\ \text{A}$ current source whose reference direction is from node 3 to 1.
- o Replace a line between the nodes 2 and 0 crossing the $30\ \Omega$ resistor by its dual $0.0333\ \text{S}$ conductor.
- o Replace a line between the nodes 2 and 4 crossing the $120\ \text{V}$ voltage source by its dual $120\ \text{A}$ current source whose reference direction is from node 2 to 4.
- o Replace a line between the nodes 3 and 0 crossing the $4\ \text{H}$ inductor by its dual $4\ \text{F}$ capacitor.
- o Replace a line between the nodes 3 and 4 crossing the $1\ \text{F}$ capacitor by its dual $1\ \text{H}$ inductor.
- o Replace a line between the nodes 4 and 0 crossing the $2\ \text{A}$ current source by its dual $2\ \text{V}$ voltage source.

The following is the construction of the dual circuit of Figure 2.

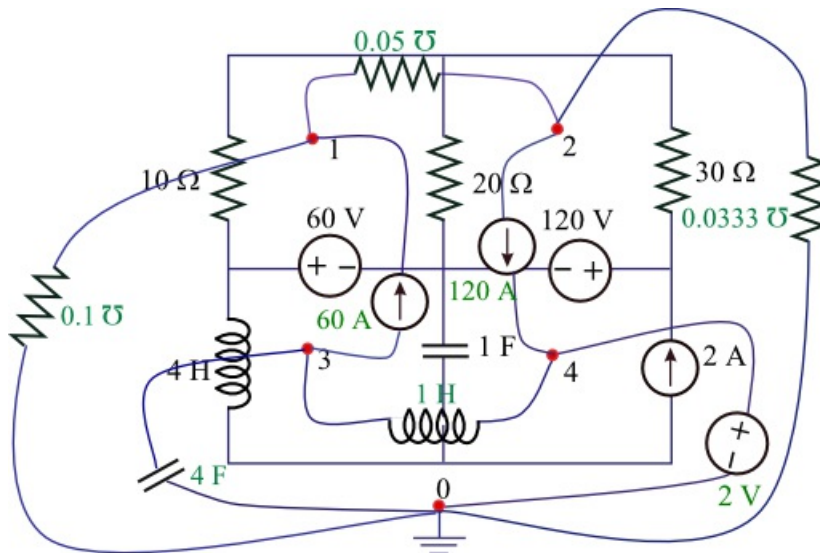


Figure 3

Step 4 of 5

Redraw the circuit in Figure 3.

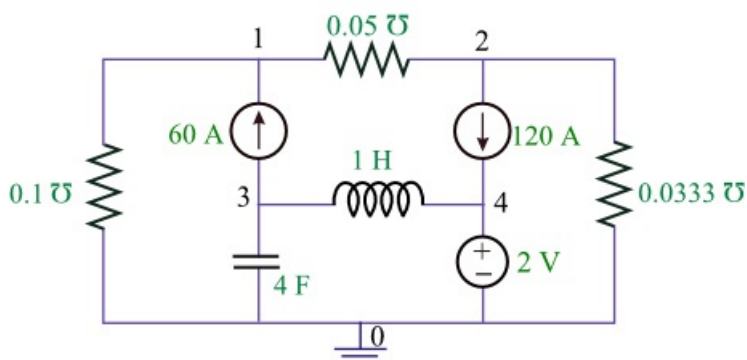


Figure 4: dual circuit for Figure 1

Thus, the circuit in Figure 4 is the dual of the circuit in Figure 1.